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Transformer guided Multimodal VQA model for Fruit recognitions

Dat Tran [a,b], Hoai Nam Vu [c,*]

[a] Thuyloi University, Hanoi, Viet Nam
[b] Graduate University of Science and Technology, Vietnam Academy of Science and Technology, Hanoi, Viet Nam
[c] Young Innovation Research Laboratory on Digital Technology (YIRLoDT), Posts and Telecommunications Institute of Technology, Hanoi, Viet Nam

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ABSTRACT

Detecting counterfeit agricultural products presents a significant challenge requiring fine-grained visual discrimination, a task where standard Visual Question Answering (VQA) models often fall short. This paper introduces a novel multi-stream, multimodal Transformer architecture designed to address this problem. Our approach features concurrent visual processing of original, segmented, and cropped image representations via a CNN-ViT pipeline enhanced with PCA-based feature transformation. Textual queries are processed using BERT, and a dedicated Fusion Transformer module deeply integrates the rich visual and textual features. This architecture is embedded within a multi-task learning framework that filters for domain-relevant questions before performing the VQA task. Evaluated on the COCO-QA benchmark and our newly created TLU-Fruit dataset, the model demonstrates superior accuracy and F1-Score against established baselines, proving particularly effective in recognizing rare agricultural products challenged by visually similar counterfeits.

1. Introduction

Visual Question Answering (VQA), a task integrating visual data with natural language queries, has advanced significantly with Large Language Models (LLMs) [1] and Transformer architectures. A critical application is in agriculture, where technology is needed to combat issues like counterfeit products, which pose substantial economic and health risks. However, agricultural VQA presents unique challenges, including the need for fine-grained visual discrimination and handling wide variations in product appearance under diverse conditions. Standard approaches often fail these specialized demands, necessitating more advanced, domain-specific models for agricultural applications [2]. Recent research has applied Transformer-based VQA in agriculture using techniques like concurrent attention structures [3] and self-attention mechanisms [4,5]. Vision-language pre-training models, such as those using frozen image encoders and LLMs [6], have also shown great potential. Despite these advancements, many agricultural applications still employ rudimentary adaptations of large models [7,8], which lack the deep, specialized feature extraction required for robust performance. While multimodal approaches generally improve accuracy [9], prediction latency remains a notable drawback, highlighting a clear research gap for an architecture that is both effective and efficient for complex agricultural VQA. To address these limitations, this paper introduces a novel multi-stream, multimodal Transformer architecture for agricultural VQA, focusing on identifying counterfeit and rare products. Our model leverages a multi-task learning framework with several key innovations: (1) concurrent visual processing of multiple image representations (original, segmented, and cropped) via a CNN-ViT pipeline with PCA enhancement; (2) BERT-based encoding for textual queries; and (3) a dedicated Fusion Transformer to deeply integrate visual and textual modalities. The primary contributions include this architecture, a new agricultural dataset named TLU-Fruit, and a comprehensive evaluation on both TLU-Fruit and the COCO-QA benchmark [10] to demonstrate our model's superior performance.

2. Materials and methods

2.1. TLU-Fruit dataset

This section outlines the collection process for the TLU-Fruit dataset, created to improve VQA system performance in answering agriculture-related queries by facilitating visual recognition and feature extraction from images combined with text (questions, answers, annotations) for better decision-making. The dataset contains 8245 question–answer pairs corresponding to 2053 images captured in diverse settings like kitchens and living rooms to provide varied environmental context. The questions and answers are categorized into two types: approximately 65% relate directly to agricultural products, aiding in their recognition, while the remaining questions are unrelated. To facilitate robust model

* Corresponding author.
E-mail addresses: dat.trananh@tlu.edu.vn (D. Tran), namvh@ptit.edu.vn (H.N. Vu).

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Table 1
Model architecture with detailed parameters and multi-task loss function.

Layer	Input	Output	Parameters
CNN (3 streams)	RGB Image (V1), Segment Image (V2), RGB Image (V3)	Flattened Feature Vectors	Conv2D (3ch in, 3 × 3 Kernel)
BERT Text Proc.	Question & Answer Tokenizers	Encoded Q & A Vectors	Embedding(30522,768), 12 Layers, Dropout(0.1)
Visual Encoder	Concatenated Features	Encoded Visual Features	BertLayer (12 Layers, 768 Hidden, GELU)
ViT Model	Image (for Patches) Patch Embeddings Encoded Sequence	Encoded Patch Embeddings Encoded Sequence (Visual Features) Pooled Features	Patch 16 × 16, Embed(3ch, 768 Hidden) 12 Transformer Layers, 768 Hidden Dense(768 Hidden, Tanh activation)
PCA Transformer	Encoded Features	Reduced Features	PCA (768-dim features)
Fusion Transformer	Reduced Features + BERT Encodings	Fused Representation	Self-Attention Layer followed by MLP (Linear(1536,512), ReLU, Dropout(0.5)), with LayerNorm
Loss Function	Multi-task Learning (Classification, Regression)		CrossEntropy + Task-specific losses
Model Settings	Text Encoder: BERT (12 Layers, 768 Hidden, 12 Attn Heads, Dropout 0.1) Image Encoder: ViT (12 Layers, Patch 16 × 16, 768 Hidden Units) PCA: Dimensionality reduction on 768-dim image features Fusion Layer: Linear(1536,512), ReLU, Dropout 0.5 Classifier: Linear(512,2), CrossEntropyLoss for classification		

Table 2
Comparison of our method and state-of-the-art methods on the COCO-QA and TLU-Fruit dataset.

Model	Coco-QA [10]			TLU-Fruit		
	Accuracy	F1-Score	W@0.9	Accuracy	F1-Score	W@0.9
DPPnet [11]	0.6119	0.6042	0.7084	0.5589	0.5537	0.5693
VinVL [12]	0.705	0.6990	0.7942	0.6398	0.6312	0.6475
Dual-MFA [13]	0.6649	0.6426	0.7615	0.5984	0.5923	0.6043
MCAN [14]	0.6808	0.6625	0.7734	0.6135	0.6175	0.6206
ODA [15]	0.6933	0.6734	0.7829	0.6212	0.6278	0.6356
v-AGCN [16]	0.7013	0.6928	0.8043	–	–	–
ALSA [17]	0.6997	0.6842	0.7943	–	–	–
MCAN+PA [18]	0.7010	0.6892	0.7943	0.6232	0.6165	0.6298
Rank VQA [10]	0.7268	**0.7153**	0.8104	0.6397	0.6313	0.6454
Ours model	**0.7364**	0.7042	**0.8295**	**0.6615**	**0.6604**	**0.6695**

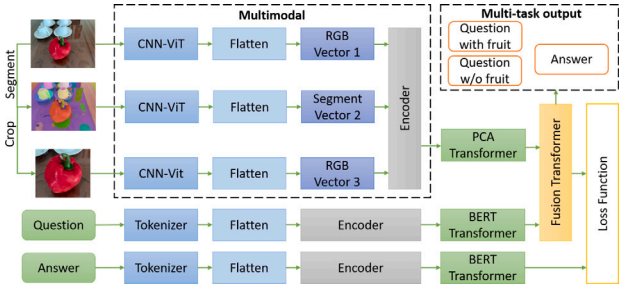

Fig. 1. Multimodal feature extractor using PCA and variations of Transformer.

development and evaluation, the TLU-Fruit dataset was subsequently partitioned at the image level–for instance, into 70% for training, 15% for validation, and 15% for testing–ensuring all question–answer pairs for a given image reside exclusively within the same set to prevent data leakage.

2.2. Model description

We propose a network architecture for VQA, detailed in Fig. 1, specifically designed to identify rare or counterfeit agricultural products by integrating multi-input CNN layers, BERT Transformer layers for text, ViT Model layers (containing Decoder and reverse Encoder networks for image embedding processing), and a PCA Transformer. As described in Table 1, the network features eight sequential neuron layers with memory units, processing information flowingly from one layer to the next. Multiple image input pathways are combined via a residual connection, flattened, and then fed into a 512-neuron fully connected layer for multitask learning. This layer first executes a detection task to filter out non-agricultural questions, focusing the system on relevant recognition tasks, and subsequently performs the answering task, generating appropriate responses (yes/no, informative, etc.) based on the agricultural query.

Specifically, our approach processes agricultural product images in multiple representations – original, segmented, and cropped – which serve as concurrent multimodal inputs alongside preprocessed textual queries to augment the model's analytical capacity. Initial visual feature extraction is performed via integrated CNN-ViT pathways. Subsequently, these visual features undergo transformation and dimensionality reduction using a Principal Component Analysis (PCA)-based module (as detailed in Table 1 and Fig. 1). This critical step not only yields a more compact and computationally efficient representation but also serves as a form of feature regularization. By projecting the features onto the principal components of maximum variance, we aim to reduce feature redundancy and filter out noise from the visual data, leading to a more robust representation for the subsequent fusion with textual features. Concurrently, textual data (questions and answers) are processed using Bidirectional Encoder Representations from Transformers (BERT) to derive their corresponding feature vectors. Our approach moves beyond simple concatenation for multimodal integration. While the refined visual features (post-PCA) and BERT-encoded textual features are initially concatenated to form a unified feature vector, the critical fusion process is performed by a dedicated module termed the 'Fusion Transformer'. This module is not a simple feed-forward network; it is a Transformer block that applies self-attention to the concatenated multimodal vector. This allows the model to dynamically assess the entire context of both the image and the text simultaneously, learning to place higher attention on the most salient features across modalities. For example, it can correlate a word like 'how many' from the question with object-specific features from the visual stream. This process models the cross-modal dependencies in a nuanced, adaptive manner, generating a deeply integrated representation that is subsequently passed to the multi-task learning heads. The combination of varied inputs – such as RGB images (providing color and structural information), segmented images (detailing shape and pixel clusters), cropped images (focusing on the specific product), and Transformer-processed text – facilitates a comprehensive semantic and contextual understanding by the model. Such an integrated framework enhances the model's ability to discern salient features, consequently improving recognition accuracy. It is crucial to note that the entire architecture, including all Transformer components, is trained end-to-end. The learning process is guided by the feedback signals from the final loss function. Gradients are backpropagated from the output layers all the

way back to the initial CNN and BERT layers, allowing the feature extractors to be fine-tuned specifically for the VQA tasks at hand. This gradient-based feedback mechanism ensures that the Transformers learn to extract and fuse the most relevant multimodal features.

Joint Learning and End-to-End Fine-tuning: A critical aspect of our architecture is that it is not a static pipeline of pre-trained blocks. While we initialize the encoders (CNN-ViT, BERT) with pre-trained weights to leverage rich, general-purpose knowledge, the entire model is subsequently fine-tuned jointly in an end-to-end fashion. This joint learning is driven by the final multi-task loss function, from which gradients are propagated back through the entire network. This process forces a co-adaptation between all modules. The visual encoders learn to produce representations that are maximally informative for the fusion module, and the fusion module, in turn, learns the optimal way to integrate these adapted features with the text. This ensures the components do not operate as disconnected parts but as a single, cohesive system optimized synergistically for the agricultural VQA task.

2.3. Loss function

To effectively guide our multi-task architecture, we employ a composite loss function, $\mathcal{L}_{\text{total}}$, which is a weighted sum of several components. Our approach focuses on the strategic combination of well-established losses to balance distinct learning objectives, rather than inventing a novel loss function. The total loss is defined in Eq. (4). Below, we justify the role of each component.

The Cross-Entropy loss is the primary engine for our classification tasks. It is applied to the output of the initial question filtering module (identifying agriculture-relevant queries) and the final answer prediction module:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{1}$$

We incorporate the Mean Squared Error loss to provide a regression signal for questions requiring numerical or continuous-valued answers, such as counting objects. This complements the classification loss by enabling the model to handle a wider variety of question types.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{2}$$

A task-specific loss for the PCA module ($\mathcal{L}_{\text{PCA}}$) can be included.

The L2 regularization term is:

$$\mathcal{L}_{\text{reg}} = \frac{\lambda}{2}\sum_{j=1}^{n} w_j^2 \tag{3}$$

The final total loss is a weighted sum of these components:

$$\mathcal{L}_{\text{total}} = \lambda_1\mathcal{L}_{\text{CE}} + \lambda_2\mathcal{L}_{\text{MSE}} + \lambda_3\mathcal{L}_{\text{PCA}} + \mathcal{L}_{\text{reg}} \tag{4}$$

where $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients, therefore critical for balancing the contributions of the primary classification task, attribute regression, and feature regularization. This composite loss function $\mathcal{L}_{\text{total}}$ acts as the central feedback mechanism for the network. By combining task-specific losses, it provides a rich and comprehensive error signal that guides the concurrent optimization of all visual, textual, and fusion modules during backpropagation.

3. Experiments

3.1. Pre-training setup

For a comprehensive comparative evaluation, Vision Transformer (ViT) and BERT [9], along with their feasibly implementable variations, were selected as baselines against our proposed model. Our experimental framework employed distinct pre-trained backbones for robust feature extraction: ResNet-101 for CNN-based image features, Facebook's Segment Anything model for generating segmentation maps, ViT for Transformer-based image processing, and BERT for text encoding. Our training strategy involves two main stages: initialization and joint fine-tuning. First, we initialize the visual and textual encoders with their public pre-trained weights. Second, the entire architecture is fine-tuned end-to-end on our target datasets. The training of our proposed model was conducted on a system equipped with an NVIDIA GeForce RTX 3090 GPU (12 GB VRAM) and 64 GB of System RAM, utilizing the AdamW optimizer ($\beta_1 = 0.8, \beta_2 = 0.96$) for up to 50 epochs with a batch size of 32. Early stopping, based on validation set performance, was incorporated to prevent overfitting. A specific learning rate schedule was implemented: 5×10^{-5}, 10×10^{-5}, and 15×10^{-5} for the first 3, 5, and 7 epochs respectively, followed by 10^{-4} for the next 10 epochs, and 10^{-6} for the remainder of the training. To ensure fair comparison and robust performance evaluation, both the collected TLU-Fruit dataset and other benchmark datasets were consistently split into training (6 parts), validation (1 part), and test (1 part) subsets, aiming for an even data distribution.

3.2. Evaluation protocol

Model performance was evaluated using several metrics to ensure robustness. Accuracy, based on the examination of answers in the training and validation sets, was calculated as the ratio of correct predictions to the total number of predictions. Additionally, the Wu-Palmer similarity (WUPS) was employed to measure the semantic correlation between predicted and ground-truth answers, utilizing concept depths and the least common subsumer (LCS) in a taxonomy:

$$\text{WUPS}(A, B) = \frac{2 \cdot \text{depth}(\text{LCS}(A, B))}{\text{depth}(A) + \text{depth}(B)} \tag{5}$$

Specific thresholds of 0.9 and 0.0 were applied (WUPS@0.9 and WUPS@0.0). Finally, the F1-score was incorporated to further enhance the evaluation's reliability and provide a comprehensive assessment of the model's predictive capabilities.

The WUPS metric was selected as a key evaluation criterion, alongside accuracy and F1-score, because it offers a more nuanced measure of correctness for VQA tasks where answers may be open-ended or allow for semantic variation. Unlike strict accuracy, which requires an exact string match, WUPS quantifies the semantic similarity between the predicted and ground-truth answers based on their positions within a common linguistic taxonomy (e.g., WordNet). This is particularly pertinent in the context of fruit recognition and agricultural VQA, where a predicted answer like "green apple" is semantically very close to a ground truth of "Granny Smith apple", or where descriptive answers about fruit condition might vary slightly in phrasing but convey the same meaning. A higher WUPS score indicates a closer semantic match. Specifically, WUPS@0.9 treats answers with a similarity score of 0.9 or higher as correct, providing a measure of near-perfect semantic agreement, while WUPS@0.0 represents the average raw similarity score across all answers, offering insight into the general semantic relatedness even for imperfect matches.

3.3. Experimental results

Our empirical study in this subsection is designed to answer the following three key research questions (RQs).

- RQ1. How do ours algorithms improve performance compared to current state-of-the-art methods?
- RQ2. What is the performance impact of ablating or simplifying key components of our proposed architecture, such as the advanced visual feature extraction pipeline (CNN-ViT with PCA enhancement), the BERT-based textual feature extraction, or the use of PCA for feature refinement?
- RQ3. How does our model perform when changes are made to the parameters of the loss function?

Table 3
Five experiments with different inputs and backbone networks.

Exp.	Image backbones (origin/seg./crop)	Text backbones (question/answer)
1	CNN	–
2	CNN-ViT	–
3	CNN-ViT	BERT
4	PCA-CNN-ViT	BERT
5	PCA-CNN-ViT	PCA-BERT

Table 4
Results for five experiments.

Experiments	Accuracy	F1-Score	W@0.9
1	0.5422	0.5486	0.5573
2	0.5734	0.5692	0.5843
3	0.6482	0.6423	0.6583
4	0.6484	0.6474	0.6529
5	**0.6615**	**0.6604**	**0.6695**

- RQ4. How can deep learning (DL) be visualized, including t-SNE plots of features and distribution plots of predicted scores from ours model?
- RQ5. How does varying the number of PCA components affect the accuracy, computational efficiency, and overall robustness of our proposed multimodal VQA model?

3.3.1. Comparison with state-of-the-art approaches (RQ1)

Comparative results on the COCO-QA [10] and the TLU-Fruit datasets, detailed in Table 2, demonstrate the effectiveness of our proposed Visual Question Answering (VQA) method. The model exhibited significant qualitative improvements, enhancing accuracy over established baselines such as BUTD, DPPnet, VinVL [12], MCAN [14], ODA, ALSA, and Rank VQA by 1%–12% on COCO-QA and 3%–11% on TLU-Fruit respectively. The efficacy of our proposed model stems from its synergistic integration of multiple models and a multitasking approach, which bolsters its capacity for joint image and text processing, proving effective in tasks like fruit counting and type identification. For instance, the model leverages attention features to accurately address queries regarding the authenticity or rarity of agricultural products, distinguishing between genuine rare items and counterfeits.

To rigorously evaluate the performance of our proposed model, statistical significance was assessed via two-sample t-tests against the RankVQA and MCAN baselines across primary metrics for both the TLU-Fruit and COCO-QA datasets. On the TLU-Fruit dataset, our model demonstrated statistically significant improvements (all $p < 0.05$) over both MCAN and RankVQA in Accuracy, F1-Score, and W@0.9. For the COCO-QA dataset, our model significantly outperformed MCAN across all key metrics (all $p < 0.01$). In comparison with RankVQA on COCO-QA, our model yielded significantly higher Accuracy and W@0.9 ($p < 0.05$). The difference in F1-Score on COCO-QA between our model (0.7042) and RankVQA (0.7153) was not found to be statistically significant. These analyses provide statistical validation for the enhanced performance achieved by our proposed architecture.

3.3.2. Applicability to fringe scenarios (RQ2)

Table 4 presents the results based on the experiments outlined in Table 3. It also demonstrates that the combination of multi-modal data yielded the best outcomes. Overall, integrating the strengths of component models, such as Transformers for images and Transformers for text, led to fairly good overall results. If only the CNN-ViT model was used, the accuracy ranged from 56% to 58%. Incorporating the BERT Transformer model further increased the accuracy by 2%–3%. When using PCA-BERT-ViT, the accuracy improved significantly, raising the average accuracy to 66%. In essence, we can conclude that adding a multi-modal approach to the VQA task provided substantial effectiveness both qualitatively and quantitatively.

Table 5
Subject independent evaluation results on the multimodal.

λ_1	λ_3	λ_3	Accuracy	F1-Score	W@0.9
0.1	0.1	0.8	0.58	0.58	0.59
0.1	0.5	0.4	0.60	0.61	0.61
0.1	0.9	–	0.62	0.62	0.63
0.5	0.1	0.4	**0.66**	**0.66**	**0.67**
0.5	0.5	–	0.63	0.63	0.64
0.5	–	0.5	0.61	0.62	0.62
0.9	0.1	–	0.63	0.64	0.64
–	0.5	0.5	0.57	0.56	0.58
–	–	1.0	0.54	0.54	0.55

3.3.3. Hyperparameter sensitivity analysis (RQ3)

As shown in Table 5, the performance of our proposed multimodal model improves significantly for the VQA task in counterfeit agricultural product identification after combining the loss functions, outperforming models that do not incorporate these loss functions. Specifically, the performance of the proposed model increases slightly as we raise the loss function parameters such as λ_1, λ_2, and λ_3. Accuracy improves by 2%–3% depending on the increase in these parameters, and it can be observed that with $\lambda_1 = 0.5$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.4$, the accuracy reaches its highest level, averaging 66%. This combination represents a fairly balanced and harmonized ratio among the loss functions, which explains the optimal accuracy achieved. In summary, we have demonstrated how the loss functions influence the proposed model and also provided parameter sets for selecting loss functions to achieve the highest possible accuracy.

In summary, this sensitivity analysis (Table 5) identifies an optimal set of weighting coefficients ($\lambda_1 = 0.5, \lambda_2 = 0.1, \lambda_3 = 0.4$) that empirically yields the highest accuracy for our model. The balanced contribution from the cross-entropy, MSE, and PCA-related loss terms with these weights appears critical for effective multi-task learning.

3.3.4. Qualitative study (RQ4)

A qualitative analysis was conducted to elucidate the model's operational characteristics and internal decision-making processes, primarily through the examination of attention matrices. These visualizations, exemplified by specific queries such as distinguishing between 'Chinese potatoes' and 'Da Lat potatoes' (a rare variety often subject to counterfeiting) as illustrated in Fig. 2, provide substantive evidence of the model's capacity to dynamically correlate salient image regions with components of the textual input. This capability is particularly evident when the model assesses fruit authenticity or type by focusing on distinct visual features. Further insights, pertinent to RQ4 and detailed through attention distributions presented in Fig. 2, reveal the mechanisms by which the model grounds textual queries in visual evidence. Specifically, it selectively attends to pertinent image regions when performing tasks such as counterfeit detection or attribute identification. This offers a significant degree of interpretability, supporting the assertion that the model engages in nuanced multimodal reasoning beyond superficial pattern matching, aligning with its intended design. Ultimately, these qualitative findings complement quantitative performance metrics, thereby providing a more holistic validation of the model's overall effectiveness.

In addition, Fig. 3 illustrates the misclassification cases in counterfeit agricultural product recognition. The image on the left in Fig. 3 shows a potato covered with soil, which affects the recognition quality. Meanwhile, the image on the right depicts strawberries on the plant, containing both ripe and unripe fruits, which also impacts recognition performance.

3.3.5. Analysis of PCA component selection (RQ5)

To justify our use of PCA and analyze the trade-off between dimensionality reduction and information loss, we performed an ablation study on the TLU-Fruit dataset. We varied the number of principal

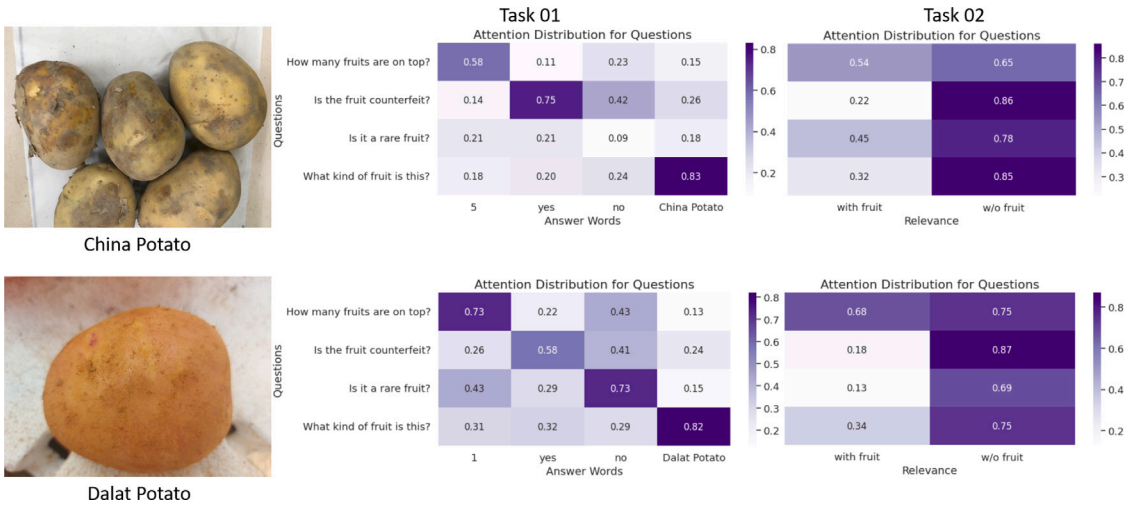

Fig. 2. Multimodal feature extractor using PCA and variations of Transformer.

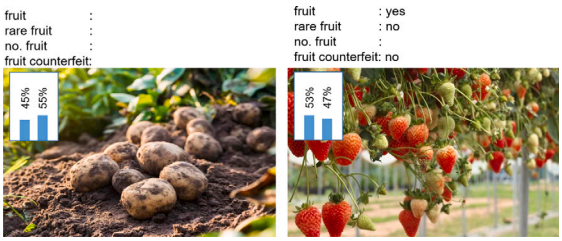

Fig. 3. While the baseline subgroup reliably classifies as 'rare fruit', it exhibits a statistically significant rise in misclassification towards 'fruit counterfeit'.

Table 6
Performance vs. number of principal components on TLU-Fruit.

Number of components	Accuracy	F1-Score
128	0.6134	0.6088
256	0.6402	0.6391
512 (proposed)	**0.6615**	**0.6604**
768 (No PCA)	0.6528	0.6501

components retained from the original 768-dimensional visual feature vectors and evaluated the model's performance. The results are presented in Table 6.

As shown in the table, retaining 512 components achieves the best performance. A lower dimensionality (e.g., 128 or 256) appears to cause significant information loss, degrading accuracy. Conversely, using the full 768-dimensional feature vector without PCA results in slightly lower performance than our chosen configuration. We attribute this to the model potentially overfitting on redundant or noisy features present in the original high-dimensional space. This analysis validates that PCA acts as an effective regularizer in our pipeline, providing an optimal balance between feature compression and model accuracy.

4. Conclusions

This paper presented an efficient multi-modal architecture that synergistically optimizes multiple pre-trained Transformer frameworks through end-to-end fine-tuning. A key component, the Fusion Transformer, serves as a deep cross-modal attention hub, effectively integrating visual and textual data to address the multi-task complexities of agricultural VQA, such as counterfeit detection. Future work will focus on enhancing practical applicability. Priorities include improving robustness against real-world imaging challenges like variable illumination and occlusions, and optimizing computational efficiency for deployment on resource-constrained edge devices. Furthermore, pilot deployments with agricultural stakeholders are planned to gather operational feedback for iterative system refinement.

CRediT authorship contribution statement

Dat Tran: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Hoai Nam Vu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] X. Ma, G. Fang, X. Wang, LLM-Pruner: On the structural pruning of large language models, Adv. Neural Inf. Process. Syst. 36 (2023) 21702–21720.
[2] B. Almadani, S.M. Mostafa, IIoT-based multimodal communication model for agriculture and agro-industries, IEEE Access 9 (2021) 10070–10088.
[3] J. Wang, et al., DualSeg: Fusing transformer and CNN structure for image segmentation in complex vineyard environment, Comput. Electron. Agric. 206 (2023) 107682.
[4] Z. Hu, et al., Prompting large language models with knowledge-injection for knowledge-based visual question answering, Big Data Min. Anal. 7 (3) (2024) 843–857.
[5] Elakkiya Rajasekar, Harshiv Chandra, Nick Pears, Subramaniyaswamy Vairavasundaram, Ketan Kotecha, Lung image quality assessment and diagnosis using generative autoencoders in unsupervised ensemble learning, Biomed. Signal Process. Control. 102 (2025) 107268.
[6] J. Li, D. Li, S. Savarese, S. Hoi, Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, in: International Conference on Machine Learning, PMLR, 2023, pp. 19730–19742.
[7] Y. Shen, L. Wang, Y. Jin, Aaformer: A multi-modal transformer network for aerial agricultural images, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2022.
[8] J. Yao, et al., Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–15.

[9] Y. Khare, et al., MMBERT: Multimodal BERT pretraining for improved medical VQA, in: Proc. IEEE Int. Symp. Biomed. Imaging, ISBI, 2021.
[10] P. Chen, et al., Enhancing visual question answering through ranking-based hybrid training and multimodal fusion, 2024, arXiv preprint arXiv:2408.07303.
[11] S. Jha, et al., A novel approach on visual question answering by parameter prediction using faster region-based convolutional neural network, Int. J. Interact. Multimed. Artif. Intell. 5 (5) (2019) 30–37.
[12] P. Zhang, et al., VinVL: Revisiting visual representations in vision-language models, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2021.
[13] P. Lu, et al., Co-attending free-form regions and detections with multi-modal multiplicative feature embedding for visual question answering, in: Proc. AAAI Conf. Artif. Intell., vol. 32, (no. 1) 2018.
[14] Z. Yu, et al., Deep modular co-attention networks for visual question answering, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2019.
[15] C. Wu, et al., Object-difference attention: A simple relational attention for visual question answering, in: Proc. ACM Int. Conf. Multimedia, 2018.
[16] J. Yu, et al., Cross-modal learning with prior visual relation knowledge, Knowl.-Based Syst. 203 (2020) 106150.
[17] Y. Liu, et al., ALSA: Adversarial learning of supervised attentions for visual question answering, IEEE Trans. Cybern. 52 (6) (2020) 4520–4533.
[18] C. Chen, D. Han, C.-C. Chang, CAAN: Context-aware attention network for visual question answering, Pattern Recognit. 132 (2022) 108980.